

INTRAVAGINAL (OR ANAL) INSERTABLE INTRALUMINAL DISTANCE/COMPLIANCE AND CONTACT DEVICE WITH CONTROLLED RADIAL STIMULUS FOR INTERNAL GEOMETRY CHARACTERIZATION (NON-LIMITING)

A. Technical Field

This section relates to insertable devices configured for intravaginal use and, in certain embodiments, anal use, to characterize internal geometry, compliance, contact integrity, and distance relationships using controlled stimulus and multi-modal sensing. The disclosed embodiments further relate to validity gating, artifact rejection, and privacy-preserving outputs suitable for standalone use and optional correlation with other nodes.

B. Summary (Non-Limiting)

In various embodiments, an insertable device is configured for intravaginal placement and, in certain embodiments, anal placement, and is used in a session-based workflow to characterize internal geometry and mechanical response. The device includes optical endpoint sensing (e.g., IR ToF), contact confirmation, and controlled radial stimulus comprising inflation/expansion elements (e.g., an annular inflatable cuff, microbladder ring, constraint band, or equivalent), with stimulus pressure measured using a MEMS pressure transducer. The device further includes compression/contact sensing using circumferential sensing (e.g., piezoresistive/FSR ring, capacitive pressure ring, piezoresistive film band), and includes a 6-axis IMU for motion/orientation sensing to improve robustness, spoof resistance, and artifact rejection. One or more temperature sensors may provide context and compensation. A computing system receives sensor signals, computes quality indices, selects valid segments, and outputs privacy-preserving internal geometry and compliance references comprising indices, zones, ranges, and confidence indicators.

C. Exemplary Physical Embodiments (Non-Limiting)

In various embodiments, the device comprises a sealed insertable body having: 1. a distal region including optical endpoint sensing for distance estimation; 2. a stimulus region comprising an expandable element configured to apply controlled radial stimulus; and 3. one or more circumferential contact sensing regions configured to detect contact distribution and response. In certain embodiments, the device includes a proximal region configured as a handle element. In certain embodiments, the device is configured with a smooth outer surface and compliant outer shell formed of silicone or other biocompatible elastomer.

D. Controlled Radial Stimulus Mechanisms (Non-Limiting)

In various embodiments, controlled radial stimulus is provided by one or more mechanisms including, without limitation: 1. an annular inflatable cuff or microbladder ring configured to expand radially; 2. a constraint band configured to shape or limit expansion; 3. manual pneumatic actuation via a compressible handle element; 4. a micro-pump configured to provide inflation control; and/or 5. a mechanical cam or equivalent mechanism configured to generate controlled expansion. In certain embodiments, the stimulus magnitude is measured by a stimulus pressure sensor including a MEMS pressure transducer and/or other pressure sensing modality.

E. Optical Endpoint and Distance Sensing (Non-Limiting)

In various embodiments, distance sensing comprises one or more optical endpoint sensors including IR time-of-flight sensing and may include multiple sensing directions to support distance estimation and internal geometry characterization. In certain embodiments, distance sensing is time-multiplexed with other sensors and validated using contact confirmation.

F. Contact Confirmation and Coupling Integrity (Non-Limiting)

In various embodiments, the device includes contact confirmation comprising one or more of: 1. capacitive proximity/contact confirmation; 2. capacitive electrodes configured to detect coupling; and/or 3. circumferential pressure/contact sensing configured to confirm tissue contact during stimulus.

G. Standalone Use and Optional Integration Hooks (Non-Limiting)

In standalone embodiments, Node 4 generates privacy-preserving internal geometry, compliance, and contact response outputs for a single user without requiring other nodes. In optional multi-node embodiments, Node 4 outputs may be time-aligned and correlated with other nodes under consent gating and quality thresholds.